

Specification-Aware Fault Diagnosis of ECG Analog Front-Ends: Bridging Alternate Test and Machine-Learning Diagnosis for In-Service Self-Test

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Abstract—An electrocardiograph whose analog front-end has drifted out of specification still produces a trace. This paper studies, by simulation, a self-test that the instrument can run in service with its own converter and internal test signals. Two questions that the literature treats separately are joined: whether the circuit still complies with the requirements of IEC 60601-2-25, as in alternate test, and which component is responsible, as in machine-learning fault diagnosis. A third one is added: whether the cause lies in the circuit or in the electrodes. About 130 000 Monte Carlo cases of two front-ends (an integrated instrumentation amplifier with its discrete network, and a discrete three-op-amp amplifier) are simulated with hard, parametric and electrode faults, gel and dry electrodes, and eleven specification values per case. On the integrated design a classifier reaches 0.9 % of escapes at 7.8 % of false rejects, against 22.5 % and 20.0 % for a limit test. Labeling a fault by component deviation instead of by loss of compliance raises false rejects from 1.6 % to 37 %. The groups of components that cannot be told apart are predicted from the sensitivities of the nominal circuit; diagnosis at that level reaches a macro F1 of 0.87 and holds for fault magnitudes absent from training, whereas component-level diagnosis does not. Non-compliant circuits are recognized in 94 % of the cases and electrode problems in 64 %. Three measurements taking 3 s decide compliance within one point of the balanced accuracy of the complete 97 s self-test.

Index Terms—Alternate test, ambiguity groups, analog fault diagnosis, biopotential amplifier, built-in self-test, electrocardiography, machine learning, skin-electrode impedance.

I. INTRODUCTION

AN electrocardiograph that has drifted out of specification still produces a trace. A gain error of ten percent, a low-frequency corner that has moved, or a degraded common-mode rejection do not stop the instrument; they change the amplitudes and the ST segment that a clinician or an algorithm will read. The required performance is laid down in IEC 60601-2-25 [1], but conformance is verified with external equipment, at type testing and at periodic maintenance. In between, the instrument cannot know whether its analog front-end still complies, and the deviations that appear in use are precisely those that the user lacks the equipment to detect [2].

What the instrument can do on its own is limited. Integrated front-ends detect a detached electrode and provide internal test signals to verify the signal chain at power-up; these are applied

inside the device and, by the manufacturer's own account, are not intended for compliance testing [3]. They do not cover the discrete network around the amplifier (protection and bias resistors, gain-setting resistor, coupling and anti-aliasing filters, driven-right-leg network, reference divider), nor say whether a deviation matters.

Two research communities have worked on parts of this problem without meeting. The analog test community checks conformance to specifications, and alternate test replaces direct specification measurements by cheap signatures from which pass or fail is inferred, with the explicit aim of neither accepting a bad circuit nor rejecting a good one [4]–[7]; it targets production test of integrated circuits. The fault-diagnosis community asks which component has failed and answers with classifiers trained on simulated faults [8]–[12]; there a fault is a component that deviates from nominal by a fixed percentage, and whether the circuit is still within specification is not asked.

For a medical instrument both questions matter, and a third appears: whether the problem is in the circuit at all, since the skin-electrode interface is part of the signal path and varies by orders of magnitude between subjects and electrode types [13]. A percentage definition of a fault is of little use here: in the front-ends studied in this paper, more than 93 % of the circuits with one component 5 % off nominal, and more than half of those with one component 50 % off, still meet every requirement of the standard. The circuit is a further issue. Chen *et al.* [2] have shown that the symmetries of the usual benchmark circuits make different faults indistinguishable, and they redesign the training circuit to remove them. An instrument designer cannot: the circuit is given. What can be done is to find out in advance which components are indistinguishable and report the diagnosis at that level.

This paper studies, by simulation, an in-service self-test of a single-lead ECG front-end that uses only what the instrument can acquire with its own converter and internal test signals. Two circuits are considered: a realistic one built around an integrated instrumentation amplifier (Circuit A) and a discrete three-op-amp amplifier that stands for the circuits of the diagnosis literature (Circuit B). The contributions are as follows.

- 1) A two-level self-test that joins specification verification and fault location for a discrete front-end network. On Circuit A, compliance with IEC 60601-2-25 is decided with 0.9 % of escapes at 7.8 % of false rejects, where a

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limit test on the same measurements gives 22.5 % and 20.0 %.

- 2) A simulation study with manufacturing tolerances of a biopotential front-end of realistic architecture: about 130 000 Monte Carlo cases with hard, parametric, amplifier and electrode faults, each labeled with eleven clinical specifications.
- 3) The separation of electrode problems from circuit faults, with gel and measured dry electrodes [1], [13], and the identification of its physical limit.
- 4) A quantified comparison of functional and percentage severity: training on component deviation instead of on compliance raises false rejects from 1.6 % to 37 %.
- 5) An open dataset and code.

In addition, the groups of components that cannot be told apart are obtained from the sensitivities of the nominal circuit, before any fault is simulated, and they explain why component-level diagnosis does not generalize to unseen fault magnitudes while group-level diagnosis does.

The study is based on simulation only, for two reasons. Labeling a case requires the eleven tests of the standard, and the campaign covers some 600 fault conditions with 200 realizations each, most of which require a component to be altered; neither is feasible on hardware at that scale. And which faults break compliance, which can be seen and which can be told apart has to be known before a bench experiment on a meaningful subset can be designed. Validation on hardware is the next step.

II. RELATED WORK

Machine-learning diagnosis: Typical studies simulate hard or soft faults under component tolerances, extract features from node voltages or from the frequency or time response, and train a classifier. Dieste-Velasco has addressed hard faults in transistor amplifier stages with a neural network on a few accessible measurements, after grouping the faults that those measurements cannot tell apart [8]; hard faults with a fuzzy inference system [9]; and soft faults in a Sallen-Key band-pass filter with seven classifiers on six voltages, reaching a classification accuracy of 97.9 % over fifteen classes [10]. Deep models on raw responses [11] and explainable classifiers on frequency-response features [12] are also used. Three aspects matter here. The Sallen-Key band-pass and the four-op-amp biquad high-pass filters recur as benchmarks [2], [10], [14], [15], and their symmetries produce overlapping fault responses [2]. Generalization across fault severities is an open problem: transferring from a 50 % to a 25 % deviation with diffusion augmentation and domain adaptation reaches 71 % on thirteen classes, a figure its authors present as an upper bound [14]. And testability theory defines ambiguity groups as sets of components whose faults cannot be distinguished from the available measurements [16]. In all this work a fault is a deviation of a component, usually 25 or 50 % against tolerances of 5 to 10 %; whether the circuit still meets its requirements is not considered.

Specification-based and alternate test: Alternate test predicts the specifications, or directly the pass/fail outcome,

from signatures that are cheap to measure; the acceptance region in signature space comes from Monte Carlo simulation and statistical classification, or a regression from signatures to specifications is calibrated on measured devices [4], [7]. Variyam and Chatterjee formulated test generation as the reduction of the probability of misclassifying a bad circuit as good and vice versa [5], [6], and redundant specification tests can be removed by statistical learning [17]. Alternate test has also served diagnosis: signatures from built-in sensors are related by statistical learning to the source of a failure in integrated RF transceivers [4]. Our work follows that principle with a different object and setting: the components of a discrete network instead of the blocks of an integrated circuit, a self-test in service with the instrument's own converter instead of production test, the requirements of a clinical standard instead of data-sheet specifications, and a confounder, the electrode, with no counterpart in integrated-circuit test.

Biomedical equipment and electrodes: Work on faults in ECG acquisition operates on the recorded signal and detects that the data are bad [18]–[20], not which part of the circuit has failed nor whether the instrument still complies. At circuit level we have found two precedents: faults of a simulated electroencephalograph circuit classified with wavelet features and a neural network, without a description of circuit or faults [21], and board-level diagnosis of a therapy device from port signals and symptoms [22]. Neither considers tolerances, specifications or electrodes. The standard represents the skin-electrode impedance by $51\text{ k}\Omega$ in parallel with 47 nF [1], whereas dry electrodes range from about $0.1\text{ M}\Omega$ to almost $9\text{ M}\Omega$, with differences of one to three orders of magnitude between subjects [13]. Such an impedance forms a divider with the input network, so a healthy instrument on a high-impedance electrode can resemble a faulty one on a good electrode, and any imbalance converts common-mode interference into a differential signal that the driven-right-leg circuit reduces but does not cancel [23]. To our knowledge no previous work separates electrode degradation from a fault of the circuit behind it.

III. METHODS

Every simulated case is one realization of a front-end: its components drawn within manufacturing tolerances, at most one injected fault, and three electrodes drawn from a population. Two simulations are run per case. The first takes the measurements that the instrument could take on itself, with the patient's electrodes connected, and gives the features. The second measures the specifications with the test networks of the standard and gives the labels.

A. Circuits

Both circuits are single-lead front-ends with the same chain: input protection and radio-frequency filtering, bias resistors, an instrumentation amplifier, a first-order high-pass filter, a gain stage, a Sallen-Key low-pass filter and a driven-right-leg (DRL) integrator.

Circuit A (Fig. 1) is built around a Texas Instruments INA333 [24] on a single 3.3 V supply; it has 16 resistors,

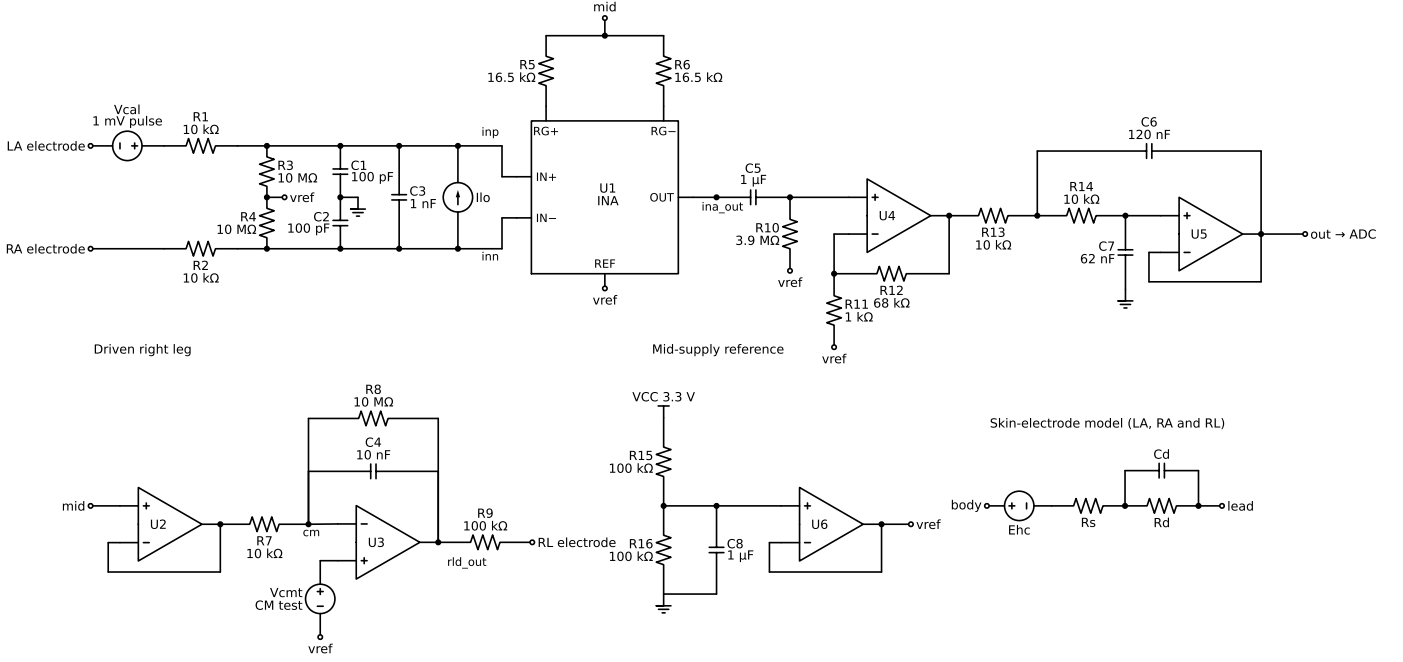


Fig. 1. Circuit A: integrated instrumentation amplifier with its discrete network, on a single 3.3 V supply. V_{cal} , V_{cmt} and I_{io} are the internal self-test sources; nets with the same name are connected. The electrode model at the lower right is placed between the body and each of the three leads.

8 capacitors and 5 discrete op-amps, and a gain of 278. Its values follow from the standard: a first-stage gain of 4.03 keeps a ± 300 mV electrode offset within the output swing; the high-pass corner is 0.041 Hz, as the impulse test requires; the low-pass filter is a Butterworth response at 185 Hz. Circuit B replaces the integrated amplifier by three op-amps and seven resistors on ± 5 V supplies (20 resistors, 6 capacitors, 6 op-amps, gain 414). It has the resistor symmetries identified in benchmark circuits [2]; its schematic is in the supplementary material.

The circuits are simulated with ngspice [25]. The INA333 is a behavioral model built from its data sheet (three-amplifier structure, single pole, noise, output clamping, offset and finite common-mode rejection ratio, CMRR), and the op-amps are single-pole models with offset, noise and clamping. Against the manufacturer's macromodel of the amplifier alone, gain and CMRR at 50 Hz agree (102 dB against 100 dB); the macromodel was not used because its fixed parameters allow neither Monte Carlo variation nor faults in the block. Healthy variation draws resistors uniformly within $\pm 1\%$ and capacitors within $\pm 5\%$, and amplifier parameters within their data-sheet ranges.

B. Electrodes

Each electrode is a half-cell potential in series with R_s and with $R_d \parallel C_d$. Half of the cases use gel electrodes, with the network of the standard (51 k Ω in parallel with 47 nF). The other half use one of six dry materials (conductive polymer, conductive fabric with and without a membrane, stainless steel, silver, platinum) with the parameters fitted in [13]; for each case one of the six subjects of the open data of that work is drawn and that subject's contact resistance is used,

which spans 0.26 M Ω to 68 M Ω for the polymer and the fabrics (referred to below as porous electrodes) and 0.6 k Ω to 0.58 M Ω for the solid metals. Every parameter of each of the three electrodes is then drawn log-uniformly between half and twice its value, so that imbalance is part of the normal variation. The patient is coupled to mains and earth as in [23]. The absence of a subject spread for gel and the factor of two between electrodes are assumptions.

C. Specifications and Labels

Eleven quantities are evaluated for every case following clause 201.12.4 of IEC 60601-2-25 [1] (Table I), with the test networks of the standard at the inputs and not the patient's electrodes. A case is *compliant* when all are within their limits. Both nominal circuits and 500 healthy realizations of each are compliant.

Each case carries three labels: the *functional* label (compliance, and the specification values), the *location* label (the component altered) and the *origin* label (none, circuit or electrode). Since the specifications belong to the circuit, an electrode fault leaves the case compliant and is recorded by its origin. The *percentage* label of the diagnosis literature is kept for comparison: a case is faulty when a component was altered.

D. Fault Model

One fault at most is injected per case (Table II). Small parametric deviations are included on purpose, since that is where functional and percentage severity disagree most. Each fault condition is simulated 200 times and the healthy circuit 5000 times, which gives 63 600 cases of Circuit A and 66 400 of Circuit B.

TABLE I
SPECIFICATIONS EVALUATED FOR EVERY CASE (IEC 60601-2-25,
CLAUSE 201.12.4) AND VALUES OF THE NOMINAL CIRCUITS

Specification	Limit	Circuit A	Circuit B
Gain error at 10 Hz	$\leq 5\%$	0	0
Response, 0.67–40 Hz, deviation	$\leq 10\%$	0.3%	0.3%
Response, 40–150 Hz, minimum	≥ 0.70	0.83	0.83
Response, 40–500 Hz, maximum	≤ 1.10	1.00	1.00
Baseline shift after impulse ^a	$\leq 100\ \mu\text{V}$	75 μV	75 μV
Baseline slope after impulse ^a	$\leq 300\ \mu\text{V/s}$	18 $\mu\text{V/s}$	18 $\mu\text{V/s}$
Common-mode rejection	$\geq 89\ \text{dB}$	122 dB	111 dB
Noise, peak to valley ^b	$\leq 30\ \mu\text{V}$	5.4 μV	3.4 μV
Loss with 620 k Ω 4.7 nF	$\leq 20\%$	3.0%	3.0%
Gain change with $\pm 300\ \text{mV}$ offset	$\leq 5\%$	0	0
Input range	$\geq 5\ \text{mV}$	5.8 mV	6.0 mV

^a3 mV, 100 ms impulse; referred to the input.

^bReferred to the input, 0.05–150 Hz; taken as 6.6 times the rms value.

TABLE II
FAULT MODEL

Fault	Model	Conditions	
		A	B
Open, short	1 G Ω in series; 1 Ω in parallel	48	52
Parametric	$\pm 5, \pm 10, \pm 20, \pm 50\%$ of nominal	192	208
Capacitor degradation	-20% with 10 Ω in series; -50% with 100 Ω	16	12
Op-amp	offset of 20 or 50 mV; open-loop gain divided by 100 or 1000	20	24
Integrated amplifier	offset of 5 or 20 mV; CMRR of 70 or 50 dB; gain error of -5 or -20%	6	–
Electrode detached	1 G Ω in series (each of three electrodes)	3	3
Electrode degraded	R_d multiplied and C_d divided by 5 or 20 (each electrode, and both input electrodes)	8	8

E. Self-Test Measurements

Three internal test sources are assumed (Fig. 1): V_{cal} in series with one lead, for a calibration pulse and a differential tone; V_{cmt} at the reference of the DRL amplifier, which the body follows, for a common-mode tone; and a differential current I_{lo} , as in ac lead-off detection. They are ideal sources. The responses are read at the output by the converter of the instrument, with the patient connected and no signal present. Four sets of measurements give 26 features:

- C1, dc levels at the output and at the instrumentation-amplifier output (2 features, 0.13 s);
- C2, gain, phase and CMRR estimate from differential and common-mode tones at 0.05, 0.5, 10, 50 and 150 Hz (15 features, 94 s);
- C3, seven descriptors of the response to a 1 mV, 200 ms calibration pulse (1 s);
- C4, response to a 1 nA lead-off current at 10 and 100 Hz (2 features, 2 s).

A measurement model is applied to the noise-free simulation results: 1 mV rms of noise at the converter input, 12 bits at 1 kHz, clipping, dc readings averaged over 64 samples and tones estimated from 1000 samples or two periods. Referred to the input of Circuit A, one code is 2.9 μV .

F. Testability Analysis

What the measurements can distinguish is analyzed before any model is trained. The sensitivity of feature f_i to component p_j of the nominal circuit is expressed in units of the spread σ_i of that feature over healthy circuits,

$$z_{ij} = \frac{0.01 p_j}{\sigma_i} \frac{\partial f_i}{\partial p_j}. \quad (1)$$

Two components whose columns of $Z = [z_{ij}]$ are parallel act on the measurements in the same way. Components whose columns have an absolute cosine of at least 0.99 are joined in an *a priori ambiguity group*; the other components and the amplifiers are groups of one. The number of singular values s of Z with $10 s \geq 3$ is reported as the testability rank. In addition, on the simulated faults, two fault conditions are called indistinguishable when no single feature separates their medians by three pooled spreads (interquartile range over 1.349); this univariate criterion is conservative and gives a map against which the models are compared.

G. Models and Evaluation

Compliance decision. Four methods use the same features. The *limit test* fails a case if any feature leaves the interval of the healthy circuits (99% joint coverage). The *specification regression* follows alternate test: one gradient-boosting regressor per specification, and a case fails if any prediction is beyond its limit. The *classifier* is a gradient-boosting classifier of compliance; with an *escape target* its threshold is lowered to the value that, on validation data, lets through 1% of the non-compliant cases.

Severity. The classifier, with class weights, is trained once on the functional and once on the percentage label, and both are scored against compliance.

Location and origin. On non-compliant cases, five classifiers (k -nearest neighbors, random forest, support vector machine, gradient boosting, multilayer perceptron) name the *a priori* group and, separately, the component. Origin is a three-class problem defined by what has to be done: *none* (healthy, or a harmless deviation), *circuit* (out of specification) and *electrode* (detached or degraded, or an open protection resistor in series with a lead, which is electrically the same as a detached electrode). Models are those of scikit-learn [26] with small hyperparameter grids.

Terminology. In the vocabulary of metrology [27], [28], the specifications are the measurands and the self-test estimates indirectly whether they are within their limits; the decision is a conformity assessment in the sense of JCGM 106 [29]. The *escape rate* is the fraction of non-conforming circuits accepted and the *false-reject rate* the fraction of conforming ones rejected. They correspond to the consumer's and the producer's risk conditioned on the true state of the circuit, over a simulated population whose proportion of non-conforming cases is set by the fault model and is not that of instruments in service; the escape target acts as a guard band. Classification accuracy, balanced accuracy, recall and F1 are figures of merit of classifiers, not statements about measurement accuracy.

Protocol. Every experiment is repeated on three random splits (52.5, 17.5 and 30% for training, validation and test),

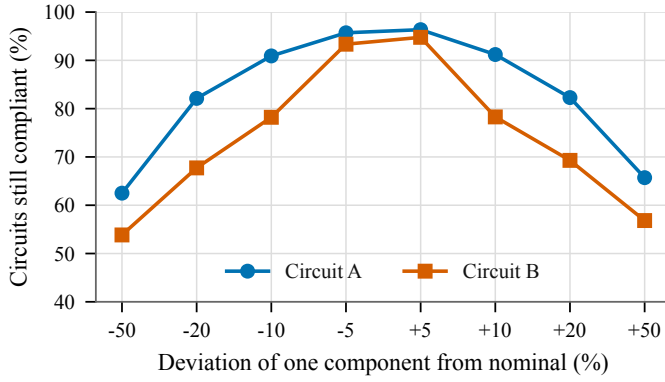


Fig. 2. Circuits that still meet every specification when one passive component deviates from nominal by the given percentage. Mean over all passive components, 200 cases each.

stratified by fault condition, with thresholds and hyperparameters chosen on validation only. Results are means over repetitions with 95 % confidence intervals (Student's t), on all electrode types and the 26 features unless stated. Robustness is examined by holding parametric magnitudes out of training, sweeping converter noise (0.1 to 10 mV) and resolution (8 to 16 bits), scoring on datasets generated with other tolerances, and changing the common-mode tone. A minimal set of measurements is found by greedy forward selection over the self-test actions.

IV. RESULTS

A. Dataset and Testability

No simulation failed. All healthy cases are compliant, and so are 79.0 % of all cases of Circuit A and 69.8 % of Circuit B, although fewer than 8 % are healthy: most injected faults do not take the circuit out of specification (Fig. 2). With one component 5 % off nominal, 93 to 96 % of the circuits remain compliant; with 50 %, 54 to 66 %. In Circuit A no fault of the DRL integrator or of its input buffer violates a specification, and neither does a CMRR of the integrated amplifier degraded to 50 dB, because the common-mode rejection of the system is dominated by the DRL path.

The electrode changes what the self-test sees. The gain that a healthy Circuit A measures through V_{cal} at 10 Hz, relative to nominal, lies between 0.90 and 1.02 with gel and solid-metal electrodes, between 0.59 and 1.00 with the polymer, and between 0.05 and 0.97 with the fabrics, whose megaohm-level impedance forms a divider with the input network. The circuit is nonetheless compliant, since the standard tests the input impedance with 620 k Ω .

Table III summarizes the analysis of Section III-F. First, some non-compliant faults cannot be seen. In Circuit A, eight of the nine invisible conditions are gain errors of 5 to 20 %, confused with the attenuation of a high-impedance electrode; without porous electrodes only the open output resistor of the DRL stage remains. In Circuit B six remain, all in the DRL stage. Second, few components can be located without ambiguity, and the groups are predictable. The sensitivities (1) alone give, for Circuit A, R5+R6+R11+R12 (the resistors that

TABLE III
TESTABILITY OF THE MAIN FEATURE SET (UNIVARIATE CRITERION)

Electrodes	Circuit A		Circuit B	
	All	No porous	All	No porous
Non-compliant conditions ^a indistinguishable from healthy	9	1	16	6
Non-compliant conditions ^a missed by the limit test	15	3	35	8
Testability rank (10 % deviation)	2	4	2	3
Components located without ambiguity	4/23	5/23	2/29	5/29
Mean number of components each one can be mistaken for	3.9	2.5	8.6	5.2

^aFault conditions in which most cases are non-compliant, out of 293 (Circuit A) and 307 (Circuit B).

TABLE IV
COMPLIANCE DECISION FROM THE MAIN FEATURE SET. MEAN OF THREE REPETITIONS AND 95 % CONFIDENCE INTERVAL, IN PERCENT

Circuit	Method	Escapes	False rejects
A	Limit test	22.5 (21.4–23.6)	20.0 (19.8–20.3)
	Specification regression	8.4 (6.7–10.2)	1.5 (1.3–1.6)
	Classifier	7.9 (6.4–9.4)	0.7 (0.4–1.0)
	Classifier, 1 % escape target	0.9 (0.8–1.0)	7.8 (4.8–10.8)
B	Limit test	35.3 (34.7–35.8)	17.0 (16.5–17.6)
	Specification regression	11.4 (10.0–12.8)	2.4 (1.9–2.9)
	Classifier	11.2 (10.3–12.1)	1.2 (1.1–1.2)
	Classifier, 1 % escape target	1.2 (0.6–1.7)	54.0 (48.0–60.0)
<i>Without porous dry electrodes</i>			
A	Classifier	4.2 (3.5–4.9)	0.7 (0.4–0.9)
	Classifier, 1 % escape target	1.0 (0.9–1.1)	1.9 (1.3–2.5)
B	Classifier	6.7 (6.2–7.1)	1.0 (0.6–1.4)
	Classifier, 1 % escape target	1.0 (0.7–1.3)	48.7 (39.4–58.1)

set the gain of the two stages), C5+R10 (high-pass filter), R13+R14 (low-pass filter) and R15+R16 (reference divider); for Circuit B, one group with the nine resistors that set the gain and three groups of two. Third, the structure depends on the architecture: Circuit A has fewer invisible faults and each component can be mistaken for fewer others.

B. Compliance Decision

Table IV and Fig. 3 give the result. The limit test accepts between a fifth and a third of the non-compliant circuits and rejects a fifth of the compliant ones: a wide healthy envelope, widened further by the electrodes, and a narrow compliance region do not coincide. The specification regression and the classifier are close to each other at their default operating points; the root-mean-square error of the predicted specifications is between 8 and 36 % of the limit, the largest for the input-impedance test.

An instrument needs a low escape rate. Lowering the threshold of the classifier brings escapes on Circuit A to 0.9 % at 7.8 % of false rejects, and at only 1.9 % without porous electrodes. On Circuit B the same escape rate costs half of the compliant circuits: the escapes that remain are the DRL faults that the measurements do not see, as the testability analysis anticipated, and no threshold recovers a fault that leaves no trace. A guard band on the regression is not a usable operating point (more than 90 % of false rejects at the same target).

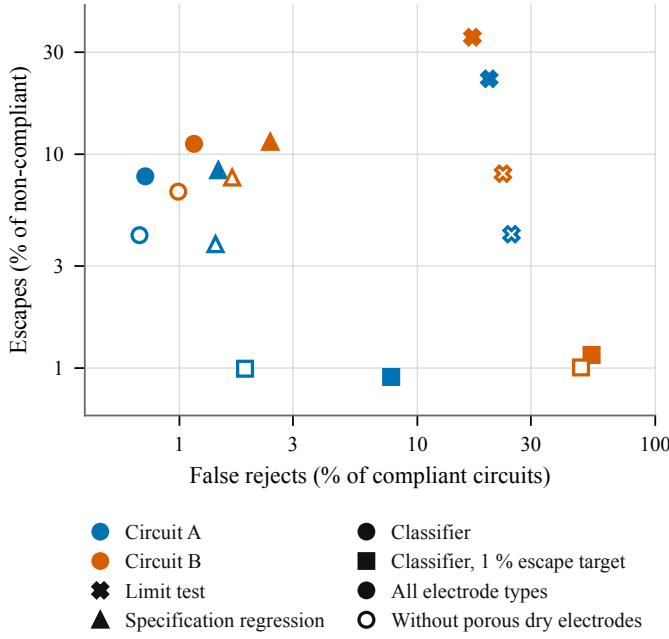


Fig. 3. Operating points of the four decision methods on the 26 features. Both axes are logarithmic; confidence intervals are in Table IV.

Telling the classifier which electrode type is connected does not help (8.3% of false rejects on Circuit A): the variation that matters is between subjects, not between materials. No single measurement set suffices; with the escape target the pulse response alone gives 44% of false rejects on Circuit A and every other set more than 75%.

C. Functional Against Percentage Severity

A detector trained to recognize that a component is out of tolerance, scored against compliance, rejects 36.8% of the compliant circuits of Circuit A and 36.5% of Circuit B, with 7.9 and 8.3% of escapes. Trained on compliance, the same model rejects 1.6 and 2.2%, with 4.1 and 7.6% of escapes. The false rejects of the percentage-trained detector are dominated by parametric faults that are real but harmless. (The escapes of the functional detector are lower than in Table IV because this experiment weights the classes, which moves the operating point.)

D. Location and Origin

There are 13364 non-compliant cases of Circuit A over 23 components and 20033 of Circuit B over 29. At the level of the a priori groups the best models reach a macro F1 of 0.87 on Circuit A (gradient boosting) and 0.76 on Circuit B (random forest), and the true group is among the three most probable in more than 99% of the cases. At component level the macro F1 falls to 0.74 and 0.61, and the errors concentrate in the predicted groups: in Circuit A the recall is 36 to 41% for the halves of the gain resistor and for the low-pass resistors, against 88 to 100% for most components outside a group. The recall of a component decreases with the number of components that the testability analysis predicted it could be

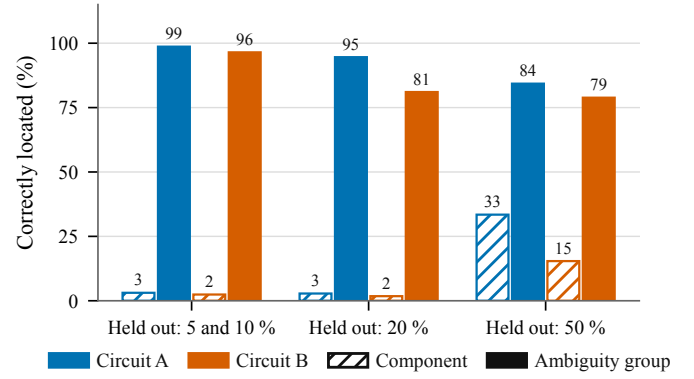


Fig. 4. Location of the cause for parametric faults whose magnitudes were held out of training. The model names the component; the group bars count a prediction as correct when it falls in the ambiguity group of the true component.

mistaken for (Spearman $\rho = -0.42$, $p = 0.045$ for Circuit A; $\rho = -0.51$, $p = 0.005$ for Circuit B). A convolutional network on the raw pulse response stays below the tabular models (0.56 and 0.32).

For origin, the macro F1 is 0.89 on Circuit A and 0.87 on Circuit B. A case with nothing to act on is recognized in 99% of the cases and a non-compliant circuit in 94 and 92%; circuit faults are almost never taken for electrode problems. Electrode problems are recognized in only 64 and 61% of the cases. An open protection resistor is always recognized and a detached electrode in 75 and 89% of the cases, but a degraded contact only in half of the cases in Circuit A and 41% in Circuit B: a contact impedance five or twenty times higher than normal for one subject falls within what healthy contacts show on other subjects. Without porous electrodes the recall of the electrode class rises to 76 and 74%.

E. Robustness

The compliance decision generalizes to fault magnitudes absent from training: its balanced accuracy falls by less than 2 points with respect to a model trained with every magnitude, except for 50% deviations on Circuit A (6 points). Component-level location does not (Fig. 4): correctly located components fall from between 21 and 70% when the magnitude was in training to between 2 and 33%. Within a group of components with parallel sensitivities the model can only tell the members apart by the size of the effect, that is, by memorizing the magnitudes it was shown. Counted at group level, the same predictions remain correct in 84 to 99% of the cases on Circuit A and 79 to 96% on Circuit B.

When the models are trained at the noise level and resolution at which they are used, results barely change from 0.1 to 10 mV of noise and from 8 to 16 bits. Trained at 1 mV and used at 10 mV, escapes on Circuit A rise from 4 to 10% and false rejects on Circuit B from 2 to 19%. A truncated Gaussian tolerance distribution changes nothing; 2% resistors and 10% capacitors raise escapes from 4.1 to 7.1% on Circuit A and from 7.6 to 11.4% on Circuit B, and leave group-level location unchanged on Circuit A (F1 0.87) but not on Circuit B (0.45). Raising the common-mode tone from 0.1 to 1 V, or making

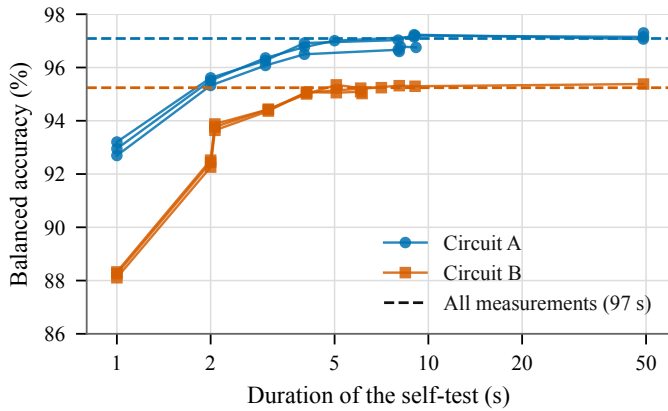


Fig. 5. Balanced accuracy of the compliance decision on the test part as self-test measurements are added one at a time by greedy selection on the validation part. One line per repetition.

it ten times longer, does not make the DRL faults detectable: about half of them escape whatever the setting. The limitation is not one of signal-to-noise ratio.

F. Minimal Set of Measurements

The calibration pulse is always selected first (Fig. 5). For the decision on Circuit A, three measurements taking 3 s (the pulse, one lead-off tone and the differential tone at 150 Hz) are within one point of the complete self-test of 97 s, in the three repetitions. Circuit B needs a fourth, a dc reading of the DRL output, which is the measurement that sees its DRL faults. For location on Circuit A, the pulse, the 150 Hz tone and the dc reading of the instrumentation-amplifier output (2 s) suffice in two of the three repetitions; Circuit B needs five or six measurements. The 0.05 Hz tones, 80 of the 97 s, are never among the first three selected.

Tables and figures for the experiments summarized in this section are in the supplementary material.

V. DISCUSSION

The results support a self-test in two levels for the realistic circuit. The compliance decision can be set to let through about 1% of the non-compliant circuits while rejecting 8% of the compliant ones, or 2% when porous dry electrodes are excluded. A limit test cannot approach this, because it asks whether the circuit looks like a healthy one, which is neither necessary nor sufficient for compliance. The second level should report an ambiguity group, not a component: the groups are known from the nominal circuit, they hold for fault magnitudes the model has not seen, and in Circuit A a group names two to four parts that set the same quantity. The comparison of labels has a direct practical reading: a detector trained in the usual way of the diagnosis literature would take out of service more than a third of the instruments that still meet the standard. The cost is low: three measurements and three seconds give the decision, and the tones below the high-pass corner are not needed, which matters because the self-test runs with the patient connected.

The electrode limits both levels. A healthy circuit on a high-impedance electrode measures a reduced gain through the calibration source, which is what a gain fault looks like; on Circuit A this accounts for eight of the nine invisible fault conditions, and false rejects fall from 7.8 to 1.9% without porous electrodes. A degraded contact can only be recognized against the normal contact of the same subject, which a single self-test does not know. A self-test that kept a reference per patient, or that could disconnect the electrodes from the calibration path, would remove most of the difficulty; neither was studied.

The two circuits differ only in how the instrumentation amplifier is built, and that changes the conclusions. The discrete amplifier adds seven resistors that all act on the gain and several DRL faults that are non-compliant and invisible; evaluated only on it, the method would have been judged unable to reach a low escape rate. This supports the argument of [2] that the circuit conditions the conclusions, with a different remedy: the ambiguity of the given circuit is computed and the diagnosis reported at that level. The invisible DRL faults also show a limit of the measurements: where a spare converter channel is available, a dc reading of the DRL output is worth more than a larger common-mode tone.

Limitations. The study is based on simulation, with behavioral amplifier models and ideal self-test sources; transfer to hardware has not been tested. Faults are single and aging is not modeled. The dry-electrode model rests on six subjects per material, and the gel model and the spread between electrodes are assumed. The noise specification is computed from the rms value and the input range from the operating point. The specifications are those of performance: some hard faults leave the circuit compliant while removing a protection (a short across the resistor that limits the current into the right-leg electrode), and a self-test aimed at safety would have to label them separately.

VI. CONCLUSION

An ECG front-end built around an integrated instrumentation amplifier can decide in service, from 26 features it measures on itself, whether it still complies with IEC 60601-2-25, with 0.9% of escapes at 7.8% of false rejects; three measurements taking three seconds come within one point of the balanced accuracy of the complete self-test. Defining a fault by its effect on the specifications, as alternate test does, instead of by the deviation of a component reduces false rejects from 37% to about 2%. The cause is located with a macro F1 of 0.87 when the answer is an ambiguity group obtained from the sensitivities of the nominal circuit, and that answer, unlike the component, remains valid for fault magnitudes absent from training. Non-compliant circuits are separated from electrode problems; degraded contacts are not, because they fall within the variation between subjects. A discrete three-op-amp front-end gives a worse picture, so conclusions drawn on benchmark-like circuits do not carry over to the circuits used in instruments. The next step is validation on hardware; multiple faults and a per-patient reference for the electrode contact are left for future work.

DATA AND CODE AVAILABILITY

The simulation pipeline, the experiments and the scripts that produce every table and figure of this paper are available at <https://github.com/telmomm/ecg-front-end-fault-diagnosis>. The dataset (both circuits, with the specification values, the three levels of labels and the configuration needed to regenerate it) is archived on Zenodo, doi: 10.5281/zenodo.23134950.

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